Digital Logic Design

Lab 08

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Experiment Title:

Design and Simulation of SOP-Based Logic Circuit, Half Adder, Full Adder, Half Subtractor, and Full Subtractor

Objectives:

- To become proficient in designing and simulating complex arithmetic and logical circuits, including Half Adders, Full Adders, Half Subtractors, and Full Subtractors.
- To simplify Boolean functions using Sum of Products (SOP) and Karnaugh Maps (K-Map).
- To implement and verify the behavior of these circuits in Proteus.

Background Theory

Arithmetic circuits are an essential part of digital systems used to perform mathematical operations like addition and subtraction on binary numbers (0s and 1s). The circuits used to carry out these operations are classified as **Adders** and **Subtractors**. In this experiment, we will explore these key arithmetic circuits along with their logic representation using Boolean functions and logic gates.

- **Adders**:
  - A **Half Adder** adds two single-bit binary numbers (A, B) and produces two outputs: **Sum (S)** and **Carry (C)**. The sum represents the basic addition result, while the carry indicates if an overflow occurred.
  - A **Full Adder** extends the Half Adder by adding three binary inputs (A, B, and a carry-in to generate a **Sum (S)** and **Carry (C)**. The full adder is vital for constructing multi-bit binary addition circuits.
- **Subtractors**:
  - A **Half Subtractor** subtracts one binary number from another (A from B), producing a **Difference (D)** and a **Borrow** output, which indicates whether borrowing is needed from the next higher bit.
  - A **Full Subtractor** builds upon the Half Subtractor by subtracting three inputs (A, B, and a borrow-in B, resulting in a **Difference (D)** and **Borrow**.

To further simplify circuit designs, Boolean functions and logical expressions are employed. These functions use **AND, OR, XOR**, and **NOT** gates to represent the relationships between inputs and outputs.

To optimize and minimize logic circuits, the **Sum of Products (SOP)** form is commonly used, which can be simplified through **Karnaugh Maps (K-Maps)**.

In this exercise, you will design a complex digital circuit in Proteus covering multiple key topics of Digital Logic Design, including Sum of Products (SOP), Karnaugh Maps (K-Map), Half-Adder, Full-Adder, Half-Subtractor, and Full-Subtractor. Follow the instructions below:

Problem Statement:

You are tasked with designing a circuit that can perform the following operations:

1.Design a combinational logic circuit using SOP that takes three inputs A, B, and C, and produces an output F that satisfies the following truth table:

Truth Table:

INPUT		**OUTPUT**	
A	**B**	**C**	**F**
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

a) Simplify the Boolean function for F using K-Map.

b) Design and simulate the circuit in Proteus using basic logic gates.

Answer: **B'C' + C'A' + AC**

2.follow the following link step by step

Design and Implement Full Adder with 3 x 8 Decoder